
ME 312P: Design Lab - 2

Course Code	ME 312P
Course Name	Design Lab - 2
L-T-P-C	0-0-2-1
Prerequisites	ME 309 Theory of Machines
Intended for	B.Tech.
Distribution	Core
One-time approval	recommended by the 9th BoA (Oct 2015)
Replaced by	ME 309P by the 61st BoA (Jan 2026)

Course Contents

- Fatigue test.
- Synthesizing a simple mechanism to produce given out put motion.
- Balancing of engines.
- 1. Study of performance of different types of governors.
2. Study of Gyroscope and precessional motion
- Generating involute profile with a rack.
- Study of gear trains and gear boxes.
- Studying follower motion with different types of cam profiles.
- Whirling speed of a shaft.
- Forced damped vibrations.
- Torsional vibrations with and without damping.
- Transmissibility of vibrations to the support.
- Use of piezo-sensors for sensing the deformation.
- Use of strain gauges, deformation of thin cylinders.
- Pressure distribution in a journal bearing.

Text Books

1. A. Ghosh, A. K. Mallik, **Theory of Mechanisms and Machines**, East West Press Pvt. Ltd.
2. Uicker, J. J., Shigley, J. E., and Pennock, G. R., **Theory of Machines and Mechanisms**, Oxford University Press.
3. Thomas Bevan, **Theory of Machines**, Pearson

References

1. C. E. Wilson, J. P. Sadler, **Kinematics and Dynamics of Machinery**, Pearson.
2. R. L. Norton, **Design of Machinery**, McGraw Hill Company